Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/05/2024
PI Name:	Costes, Sylvain Ph.D.		
Project Title:	Mapping Peripheral Immune Signatures of Mouse and Human Responses to Space Radiation for Biomarker Identification		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	sylvain.costes@bmsis.org	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	650-604-5343
Organization Name:	Blue Marble Space Institute of Science		
PI Address 1:	600 1st Avenue, 1st Floor		
PI Address 2:			
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98104	**Congressional District:**	7
Comments:	NOTE: After retiring from NASA, the PI joined Blue Marble. Dr. Costes was previously at Lawrence Berkeley National Laboratory until December 2016.		
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	10/01/2020	**End Date:**	09/30/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2024 per E. Cekanaviciute (Ed., 8/1/24) NOTE: End date changed to 09/30/2023 per PI (Ed., 10/13/22)		
Key Personnel Changes/Previous PI:	CoInvestigator Susanna Rosi, Ph.D. has left the project. Sonali Verma (Research Associate at Blue Marble Space Institute of Science) left in June 2022. Dr. Connie Tsai Pasternak, Ph.D. (Staff Scientist at Blue Marble Space Institute of Science) left October 2022. Dr. Kira Rienecker, Ph.D. (Staff Scientist at Blue Marble Space Institute of Science) joined the project, replacing Dr. Connie Tsai Pasternak in October 2022 and left on Aug 2023 Gabriela Rincon, NASA Office of Science, Technology, Engineering and Mathematics (OSTEM) Engagement intern joined the project from September - December 2022. and left on Jan 2023 Bansal, Ananya (Research Associate at Blue Marble Space Institute of Science), joined the project in Jan 2023 and left in June 2024		
COI Name (Institution):	Cekanaviciute, Egle Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Deep space exploration, including upcoming lunar and Mars missions will require repeated, real-time, minimally invasive astronaut monitoring to identify and mitigate the health impairments that are caused by prolonged exposure to the space environment. Immune and central nervous system (CNS) dysfunctions are among the main health risks associated with spaceflight, and they are predicted to be exacerbated by deep space radiation, including galactic cosmic rays (GCRs), thus, there exists an unmet need to develop biomarkers that would be applicable to humans in addition to model organisms. We propose to identify a signature of peripheral blood-based biomarkers of responses to GCRs and combined exposures including microgravity and social isolation, spanning mouse models and human subjects. We aim to develop a multi-factorial model that might subsequently be used to identify the most robust potential biomarkers or their combinations for use in spaceflight. We will combine biomarkers that had been extensively studied in the context of ionizing radiation, such as DNA damage and repair, with novel approaches, including the amount and content of exosomes. We will acquire and share samples that will be collected as part of existing and upcoming studies. We will analyze mouse peripheral blood mononuclear cells (PBMCs) isolated after GCR irradiation and combined exposures to microgravity and social isolation, as well as cultured human PBMCs responses to GCR components ex vivo. To establish the relative biological efficacy of GCRs we will supplement these studies with mouse and human PBMCs exposed to both high and low dose rate gamma irradiation. We propose the following metrics, all of which can be obtained within a comparably short time from a peripheral blood draw and thus might serve as biomarkers in spaceflight: a) DNA damage kinetics, b) oxidative stress, c) cell death, and d) exosomal quantification and characteristics. All outcomes will be evaluated in correlation with the severity of behavioral and physiological CNS deficits and analyzed together with other peripheral and immune biomarkers already available from the same samples. All results will be compared at different time points and after different doses of irradiation to generate a more complete representation of their responses. All samples will be obtained from male and female individuals to account for sexual dimorphism. By the end of this project, we anticipate to be able to identify a combination of blood-based biomarkers that robustly and reproducibly respond to GCR exposures, correlate with physiological and behavioral impairments, and are present in both humans and animal models. We will also compare them to the effects of gamma irradiation, assess their dependence on other spaceflight stressors, and evaluate sex-dependent differences in responses. Our research might be used in the future to validate the key pathways and outcomes across multiple spaceflight and ground studies to improve the health risk prediction for astronauts on deep space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work will help develop biomarkers for personalized risk assessment for therapeutic radiation in addition to space radiation.
Task Progress:	Prior to this report, we had finished the quantification of ex vivo human immune cell radiosensitivity to low- and high-LET particle and photon irradiation, with the focus on CD4+ T helper lymphocytes, CD8+ cytotoxic T lymphocytes, CD11b+ monocytes, and CD19+ B lymphocytes. In the past year, we have significantly expanded the study by adding more human subject samples to analyze demographic associations with immune cell distribution and responses to different time points after irradiation and in response to different radiation qualities, as demonstrated both by DNA damage and repair and by oxidative stress markers and antioxidant production. Furthermore, we have performed a comprehensive analysis of immune cell-type dependent radiosensitivity to photon and particle radiation based on 53BP1+ DNA repair foci kinetics and cell loss. We have dedicated a major effort towards quantifying ex vivo and in vivo mouse immune cell radiosensitivity to the same qualities and doses of ionizing radiation; however, the samples acquired as part of the tissue sharing project failed to respond to our standard immunostaining techniques used on the same type of samples in the lab or to a range of procedures available in literature. Therefore, we are planning to focus on human immune radiosensitivity in the upcoming publication. Finally, we have collected experimental immunostaining data for developing an AI/ML tool for human immune cell type categorization for further studies. Our work on this project has been presented at the NASA Human Research Program Investigators' Workshop and the Committee on Space Research (COSPAR) international meeting and has been submitted for the American Society for Gravitational and Space Research (ASGSR) annual meeting.
Bibliography Type:	Description: (Last Updated: 06/03/2026)